ConvoShop: Technologies Used

ConvoShop is an intent-driven e-commerce product advisory app (a retrieval-augmented generation framework). The project has a React frontend and a Python data and AI pipeline.

Frontend (Frontend/my-react-app)

Technology	What we use it for
React 19	Builds the whole UI from components, using hooks (useState, useEffect, useMemo, useRef)
Vite 8	Dev server (npm run dev) and production build (npm run build)
@vitejs/plugin-react	Lets Vite compile JSX and gives instant refresh while coding
PapaParse 5	Reads the Amazon products CSV (about 42,000 rows) in the browser
Plain CSS	All styling (App.css, index.css), with no UI library
Google Fonts	Typefaces: Bricolage Grotesque (headings) and Instrument Sans (body)
ESLint 10	Checks code quality, including the React hooks rules
localStorage / sessionStorage	Keeps the demo login, saved users and each user's cart in the browser

There is no backend yet. Products come from a CSV file and accounts live only in the browser, which is fine for a demo.

Data and AI pipeline (Phone_Datasets)

Technology	What we use it for
Python	Language for all data scripts, training and knowledge-base building
pandas, numpy, pyarrow	Cleaning, merging and storing the phone and product data
scikit-learn	Training the intent model
sentence-transformers	Creating text embeddings, including fine-tuning
FAISS (faiss-cpu), ChromaDB	Vector search over the knowledge base (the retrieval part of RAG)
Kaggle API, Hugging Face Hub	Downloading the source datasets
Jupyter notebooks	Data exploration, cleaning and knowledge-base building
PyYAML, python-dotenv	Config file and environment settings
pytest	Automated tests for the cleaning code

Datasets: the Amazon products sales CSV (about 42,000 products) is used by the frontend. Phone specs and reviews from Kaggle and Hugging Face are used in the data pipeline. The frontend and the pipeline are not connected yet.

Frontend folder guide

```
Frontend/my-react-app/
|-- index.html
|-- package.json
|-- vite.config.js
|-- eslint.config.js
|-- public/
|   |-- amazon_products_sales_data_cleaned.csv
|   |-- logo.png, favicon.svg, icons.svg
|-- src/
|   |-- main.jsx
|   |-- App.jsx
|   |-- App.css
|   |-- index.css
|   |-- assets/
|-- node_modules/
```

Path	What it is for
index.html	The single HTML page. It has an empty div with id "root" where React mounts the app.
package.json	Lists the dependencies and the scripts (dev, build, lint, preview).
vite.config.js	Vite settings, such as enabling the React plugin.
eslint.config.js	Linting rules for the project.
public/	Static files served as they are: the product CSV, logo.png, favicon.svg and icons.svg.
src/	All the application source code.
src/main.jsx	Entry point. Renders the App component into the page and loads index.css.
src/App.jsx	The main file: Login screen, Convo advisor, product grid, product popup and cart.
src/App.css	Styles for the whole app: header, hero, cards, popup, cart drawer, login.
src/index.css	A small global reset.
src/assets/	Images imported by the code (react.svg, vite.svg, hero.png).
node_modules/	Installed packages, created by npm install. Never commit or share this folder.

How the app works

1. main.jsx starts React and mounts App.

2. App shows the Login screen until a user signs in. The session is kept in the browser.

3. After login, the Store loads the CSV with PapaParse and builds the product list.

4. Convo, the robot in the hero, reads a plain-English request such as "a phone under $300", ranks the matching products and explains each pick.

5. The product grid has category chips, search, sorting and infinite scroll. Products open in a popup, and the cart is saved per user in localStorage.

Run it locally

```
cd Frontend/my-react-app
npm install
npm run dev
```

Then open http://localhost:5173 in the browser.